Beijing-Dublin International College

COMP3040J - Advanced S/W Technologies 2
Assignment 1

Assignment Details:

- **Assignment Type:** Code (Jupyter Notebook)
- **Release Date:** 20-April-2026
- **Lecturer:** Helard Becerra
- **Weighting:** 15%
- **Due Date:** 11-May-2025 (3 Weeks)
- **Method of Submission:** Moodle (.zip file containing .ipynb file plus .wav recordings)
- **AI engagement**: Level 3 - AI Collaboration

Assignment Description:

This assignment introduces students to the analysis of speech signals using signal-processing techniques and feature extraction methods. You will analyse natural and synthetic speech recordings, compute acoustic features, and interpret the results in relation to concepts covered in lectures. The assignment consists of **four parts**. Use the provided **Jupyter Notebook .ipynb file** to include your python code and your answers. **No other code file is needed for this assignment.**

Deliverables:

Submit a .zip file using Moodle. The zip file should contain:

- Jupyter Notebook: .ipynb file containing the code implementation and written answers.
- `student` folder containing 11 recorded .wav files
- Three (3) concatenated .wav files

Grading:

- Part 1: Speech Recording and Data Preparation **20%**
- Part 2: Formant Extraction with AI Assistance **20%**
- Part 3: Formant Space Visualization **20%**
- Part 4: Pitch Estimation and Spectrogram Analysis **20%**
- Code readability, organization, documentation **20%**

Part 1: Speech Recording and Data Preparation

1. Create **your own set of 11 spoken words**. Record yourself pronouncing each of these words clearly:
   - hid
   - hood
   - head
   - whod
   - heed
   - hudd
   - hod
   - had
   - heard
   - hard
   - hoard
2. For each word:
   - Save the recording as an individual **.wav file**.
   - Ensure that each recording contains only **one spoken word**.
   - Ensure the recording is **Mono** and has a sampling rate of **44.10 kHz**
   - Follow this filename format: `<YOUR-UCD-STUDENT-ID>-<WORD>.wav`. Example: `45452323-hard.wav`
3. Create a folder named: **`student`**. Place your **11 recorded wav files** in this folder. Additionally, you are provided with two sets of speech recordings:
   - Folder synthetic1
     - Files follow the format: `syn1-<WORD>.wav`
   - Folder synthetic2
     - Files follow the format: `syn2-<WORD>.wav`
4. Using the provided **Jupyter Notebook template**, load all wav files from the three folders: `student`, `synthetic1`, `synthetic2`.
5. Create a pandas DataFrame with the following columns: `["speaker", "word", "F1", "F2"]`. Populate the following columns:
   - **speaker:** student, synthetic1, or synthetic2
   - **word:** the spoken word extracted from the filename
   - Leave **F1** and **F2** empty at this stage.

Part 2: Formant Extraction with AI Assistance

This assignment follows a Level 3 AI Engagement Policy (AI Collaboration). Students are permitted to use Generative AI tools to assist with technical implementation, but must clearly document their use.

1. Using an AI tool of your choice, compute the **first two formant frequencies (F1 and F2)** for all speech recordings. Update the DataFrame columns: **F1** and **F2** with the computed values.

2. In your notebook, include the following:

   - **The AI tool used**
   - **Prompts used** Include the exact prompts you submitted to the AI tool and the responses obtained.
   - **Methods and libraries used** Identify the signal-processing methods and Python libraries used.
   - **Connection to lecture material** Explain how the formant extraction methods relate to the concepts discussed in lectures
   - **Prompt modification or correction** If you modified the AI-generated solution, explain what changes were made, and explain why the modification was necessary.

Part 3: Formant Space Visualization

1. Using the information stored in your DataFrame, generate a scatter plot using the following axes:
   - x-axis: F1
   - y-axis: -(F2 - F1)
2. Plot the results for the three sets of recordings: student, synthetic1 and synthetic2. For the plot, follow these requirements:
   - Use different colours for each speaker category.
   - Add text annotations indicating the word at each plotted point.
3. Answer the following questions based on the plot:
   - Do the points corresponding to multiple recordings of the same word cluster together in the formant space? - Explain your observations.
   - Do the synthetic speech recordings cluster together with the natural (student) recordings, or do they appear in different regions? - Explain your observations.

Part 4: Pitch Estimation and Spectrogram Analysis

For each set of recordings (student, synthetic1, synthetic2):

1. Concatenate the words into a single audio signal per set. This will result in three combined audio wav files.
2. Using the Python library librosa, apply the pyin algorithm to estimate the pitch (fundamental frequency, F0) for the three wav files.
3. Generate the following visualisation:
   - Spectrogram of the concatenated signal
   - The estimated pitch contour (F0) on the spectrogram.